Exact higher moments of the limiting spectral distributions of random Toeplitz and Hankel matrices

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Abstract

The limiting spectral distributions γ_T and γ_H of random symmetric Toeplitz and Hankel matrices (Bryc–Dembo–Jiang, *Ann. Probab.* 34 (2006); Hammond–Miller, *J. Theoret. Probab.* 18 (2005)) have been known to exist for two decades, yet remain unidentified; their explicitly published moments stop at order eight. We compute the moments exactly through order 14 for both ensembles, in exact rational arithmetic: $m_{10}(\gamma_T) = 415$, $m_{12}(\gamma_T) = 23840/7$, $m_{14}(\gamma_T) = 325719/10$, $m_{10}(\gamma_H) = 2717/36$, $m_{12}(\gamma_H) = 1052/3$, $m_{14}(\gamma_H) = 1331087/720$. The computation also yields $m_8(\gamma_T) = 908/15$, correcting the value 964/15 reported by Hammond–Miller. Each moment is a sum of volumes of explicit rational polytopes indexed by pair partitions; we evaluate every volume exactly by a double-description/triangulation engine and check the results against an independent vertex enumeration, Monte Carlo integration, direct eigenvalue simulation, an exact counting route, and all previously published values. We further prove a sharp structural theorem: a pairing has volume 1 if and only if it is non-crossing, and every crossing pairing has volume at most 2/3 — attained in the Toeplitz case — in both ensembles; consequently $m_{2k}(\gamma_T) \le \mathrm{Cat}_k + \frac{2}{3}\big((2k-1)!! - \mathrm{Cat}_k\big)$ and $m_{2k}(\gamma_H) \le \mathrm{Cat}_k + \frac{2}{3}\big(k! - \mathrm{Cat}_k\big)$. The complete volume spectra are tabulated, in the paper and in the accompanying repository; they are supported on a small set of rational values, shared across the two ensembles. We close with two numerical observations: the Gaussian-deficit ratios of γ_T trend toward the square of the Sen–Virág spectral-norm constant, and the analogous Hankel ratios suggest a value for the (open) Hankel norm constant.

1 Introduction

Let $\{a_j\}_{j\ge 0}$ be i.i.d. real random variables with $\mathbb{E}a_0 = 0$, $\mathbb{E}a_0^2 = 1$ and all moments finite. The random symmetric Toeplitz and Hankel matrices are

$$T_n = \big(a_{|i-j|}\big)_{i,j=0}^{n-1}, \qquad H_n = \big(a_{i+j}\big)_{i,j=0}^{n-1}.$$

Bryc, Dembo and Jiang [1] and Hammond and Miller [2] proved that the empirical spectral distributions of $T_n/\sqrt{n}$ and $H_n/\sqrt{n}$ converge weakly, almost surely, to nonrandom symmetric probability measures γ_T, γ_H that do not depend on the entry law. Both measures have unbounded support; γ_H is not unimodal [1]; γ_T is absolutely continuous with bounded density

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[8]; the spectral norm of T_n obeys $\|T_n\|/\sqrt{2n\log n} \to 0.8288\ldots$ [7, 9]. Identifying γ_T or γ_H in closed form is a recognized open problem: already the 2003 review [6] noted of the then-new existence result that "its closed form is not yet known", and two decades later the laws remain unidentified. Recent work provides alternative *representations* of the limits — moment formulas for dependent-entry generalizations that include the i.i.d. case [4], and a description via $*$-convergence of circulant-type matrices [5] — without producing closed forms or explicit higher moments. The only explicitly published moment values are

$$m_4(\gamma_T) = \tfrac{8}{3}, \quad m_6(\gamma_T) = 11, \quad m_8(\gamma_T) \text{ (see below)}, \qquad m_4(\gamma_H) = 2, \quad m_6(\gamma_H) = \tfrac{11}{2}, \quad m_8(\gamma_H) = \tfrac{281}{15}.$$

Since both limit laws are symmetric and determined by their moments [1], exact higher moments are the natural invariants through which these unidentified laws can be studied, compared against candidate closed forms, and reconstructed numerically. This paper contributes:

1. **New exact moments** (Theorem 5.1): $m_{10}(\gamma_T) = 415$, $m_{12}(\gamma_T) = \frac{23840}{7}$, $m_{14}(\gamma_T) = \frac{325719}{10}$, $m_{10}(\gamma_H) = \frac{2717}{36}$, $m_{12}(\gamma_H) = \frac{1052}{3}$, $m_{14}(\gamma_H) = \frac{1331087}{720}$, all in exact rational arithmetic, with the complete volume spectrum of every order tabulated (Section 5 and the data repository).

2. **A correction**: the eighth Toeplitz moment is $m_8(\gamma_T) = \frac{908}{15} = 60.5\overline{3}$, while [2] reports $64\frac{4}{15} = \frac{964}{15}$; Section 5.2 gives the configuration count behind the corrected value, in the format of [2], together with the verifications.

3. **A sharp structural theorem** (Theorem 3.5): in both ensembles the polytope volume attached to a pair partition equals 1 exactly when the pairing is non-crossing, and is at most 2/3 otherwise — with 2/3 attained (Toeplitz). This yields explicit upper bounds for *all* moments (Corollary 3.6) and explains the leading structure of the moment sequences.

4. **Quantitative consequences** (Section 6): explicit tail bounds; Gaussian-deficit sequences; and two numerical observations, connecting the deficit ratios of γ_T to the Sen–Virág constant 0.8289 and suggesting an estimate for the unknown Hankel norm constant.

2 The moment–volume formulas

We recall and re-derive the volume representation of the limit moments, in the precise normalization used by our computations. Let $\mathcal{P}_2(2k)$ denote the perfect matchings (*pairings*) of $\{0, 1, \ldots, 2k-1\}$; $|\mathcal{P}_2(2k)| = (2k-1)!!$. For $\pi \in \mathcal{P}_2(2k)$ write $\pi = \{\{l_1, m_1\}, \ldots, \{l_k, m_k\}\}$ with $l_c < m_c$, let $v(r)$ be the index of the pair containing slot r, and say π is *parity-mixing* if every pair joins an even and an odd slot.

Theorem 2.1 (essentially [1, 2])**.** *For $t = (t_1, \ldots, t_k) \in \mathbb{R}^k$ define $S_l(t) = \sum_{r<l} \sigma(r)\, t_{v(r)}$, where $\sigma(r) = +1$ if r is the smaller element of its pair and -1 otherwise. Let*

$$P_\pi = \Big\{(x_0, t) \in \mathbb{R}^{k+1} :\ 0 \le x_0 + S_l(t) \le 1,\ l = 0, 1, \ldots, 2k-1\Big\}, \qquad p_T(\pi) = \mathrm{vol}_{k+1}(P_\pi).$$

Then $m_{2k}(\gamma_T) = \sum_{\pi \in \mathcal{P}_2(2k)} p_T(\pi)$.

For parity-mixing π define $A_l(x_0, y) = (-1)^l x_0 + \sum_{r<l} (-1)^{l-1-r} y_{v(r)}$ and

$$P_\pi^H = \Big\{ (x_0, y) \in \mathbb{R}^{k+1} : \ 0 \le A_l(x_0, y) \le 1, \ l = 0, \dots, 2k-1 \Big\}, \qquad p_H(\pi) = \mathrm{vol}_{k+1}(P_\pi^H).$$

Then $m_{2k}(\gamma_H) = \sum_{\pi \text{ parity-mixing}} p_H(\pi)$, the sum having $k!$ terms.

Proof. Expanding $\mathbb{E}\,\mathrm{tr}\, T_n^{2k}$ over closed index circuits $(i_0, \dots, i_{2k-1})$, the slot values are $d_l = i_{l+1} - i_l$ (indices mod $2k$), $\sum_l d_l = 0$. Independence and centering force every value of $|d_l|$ to occur at least twice; configurations with fewer than k distinct values are $O(n^k)$ in number against the normalization $n^{-(k+1)}$, hence vanish in the limit, leaving pair partitions with $\mathbb{E}a^2 = 1$ per pair. For a matched pair $\{l, m\}$, $d_m = \pm d_l$; if $d_m = +d_l$ for some pair, the closure $\sum_l d_l = 0$ becomes a nontrivial linear constraint on the k free differences, again $O(n^k)$. So at leading order $d_{m_c} = -d_{l_c}$ for all pairs, and the circuit is determined by i_0 and the k values $x_c := d_{l_c}$, subject to $i_l = i_0 + \sum_{r<l} \sigma(r) x_{v(r)} \in [0, n)$ for all l. Scaling $x_0 = i_0/n$, $t_c = x_c/n$ and letting $n \to \infty$, the normalized count is a Riemann sum converging to $\mathrm{vol}(P_\pi)$ (the constraints $t_c \in [-1, 1]$ being implied by consecutive slot constraints). Boundary strata ($x_c = 0$, ties $|x_c| = |x_{c'}|$, diagonal entries) have codimension ≥ 1 and do not contribute. Convergence of $\mathbb{E}$-moments plus the concentration proved in [1] upgrades this to the moments of γ_T.

For Hankel the slot values are $s_l = i_l + i_{l+1}$, whence $i_l = (-1)^l i_0 + \sum_{r<l} (-1)^{l-1-r} s_r$ and the closure constraint $\sum_r (-1)^r s_r = 0$. Matching s-values in pairs, the closure is automatic precisely for parity-mixing pairings and a nontrivial constraint otherwise (one degree of freedom lost, hence negligible). Scaling $y_c = s_{l_c}/n \in [0, 2]$ gives the stated polytope; the bounds $y_c \in [0, 2]$ are implied by consecutive constraints. □

Remark 2.2. P_π is cut out by at most $4k$ halfspaces with coefficients in $\{-1, 0, 1\}$ and right-hand sides in $\{0, 1\}$, so $p_T(\pi), p_H(\pi) \in \mathbb{Q}$. The first values: $m_2 = 1$ for both ensembles; $m_4(\gamma_T) = 1 + 1 + \frac{2}{3} = \frac{8}{3}$; $m_4(\gamma_H) = 1 + 1 = 2$.

2.1 Symmetries and reductions

Lemma 2.3 (dihedral invariance)**.** *p_T and p_H are constant on orbits of the dihedral group D_{2k} acting on slots (rotations $r \mapsto r+1$ and reflections $r \mapsto c - r$ mod $2k$; for Hankel both operations preserve the parity-mixing class).*

Proof. The set of admissible circuits $\{(i_0, \dots, i_{2k-1})\}$ counted in the proof of Theorem 2.1 is intrinsic to the cyclic word $a_{w(i_0,i_1)} a_{w(i_1,i_2)} \cdots a_{w(i_{2k-1},i_0)}$ ($w(i,j) = |i-j|$ resp. $i+j$): relabeling the circuit cyclically or reversing its orientation maps admissible circuits for π bijectively onto those for the transformed pairing. The volume is the $n \to \infty$ normalized count, hence equal. □

Lemma 2.4 (adjacent-pair deletion)**.** *If π pairs two adjacent slots $\{l, l+1\}$, deleting them (and relabeling) yields $\pi' \in \mathcal{P}_2(2k-2)$ with $p(\pi) = p(\pi')$, in both ensembles.*

Proof. The pair's variable appears in exactly one constraint, $0 \le x_0 + S_l \pm t \le 1$ (resp. $0 \le y - A_l \le 1$), and in no other; given the remaining variables the constraint confines it to an interval of length exactly 1 contained in its implied range. Fubini integrates it out with factor 1. □

Since every non-crossing pairing contains an adjacent pair, induction gives:

Corollary 2.5. *$p_T(\pi) = p_H(\pi) = 1$ for every non-crossing π. (Every non-crossing pairing is parity-mixing, so all $\mathrm{Cat}_k = \binom{2k}{k}/(k+1)$ of them appear in both ensembles.)*

3 Volume 1 characterizes non-crossing pairings, with a $2/3$ gap

Throughout this section fix $\pi \in \mathcal{P}_2(2k)$ with pairs $\{l_c, m_c\}$, $l_c < m_c$, labelled so that $l_1 < l_2 < \cdots < l_k$. Write

$$O(l) := \{c : \ l_c < l \le m_c\} \qquad \text{(the pairs } \textit{open} \text{ at slot } l), \qquad l = 0, 1, \ldots, 2k.$$

Lemma 3.1 (opening-slot coordinates, Toeplitz)**.** *Let $W_l = x_0 + S_l(t)$ for $l = 0, \ldots, 2k-1$, so that $P_\pi = \{(x_0, t) : W_l \in [0,1] \ \forall l\}$. Then: (i) $W_l = x_0 + \sum_{c \in O(l)} t_c$ for every l; (ii) the linear map $\Phi : (x_0, t_1, \ldots, t_k) \mapsto (W_0, W_{l_1+1}, \ldots, W_{l_k+1})$ is unimodular, i.e. $\det \Phi = \pm 1$; (iii) every W_l expands uniquely as an integer combination of the* basis *coordinates $\Phi_0 := W_0$, $\Phi_c := W_{l_c+1}$ $(1 \le c \le k)$, with no constant term, and the coefficients (α_i) of any such expansion satisfy $\sum_i \alpha_i = 1$.*

Proof. (i) In $S_l = \sum_{r<l} \sigma(r) t_{v(r)}$, a pair c with $m_c < l$ contributes $t_c - t_c = 0$, a pair with $l_c < l \le m_c$ contributes $+t_c$, and later pairs contribute nothing. (ii) $W_0 = x_0$ and $W_{l_c+1} = x_0 + t_c + \sum_{c' \in O(l_c+1),\, c' \ne c} t_{c'}$, where every $c' \in O(l_c+1) \setminus \{c\}$ has $l_{c'} < l_c$; in the variable order $(x_0, t_1, \ldots, t_k)$ the matrix of Φ is lower triangular with unit diagonal. (iii) Unimodularity gives unique integral expansions; all the forms involved are homogeneous linear in (x_0, t), so no constant appears, and comparing the coefficient of x_0 on both sides of $W_l = \sum_i \alpha_i \Phi_i$ (each Φ_i has x_0-coefficient 1, as does W_l) gives $\sum_i \alpha_i = 1$. □

Lemma 3.2 (crossings force a thick slot)**.** *If π has a crossing, some slot $l^* \in \{1, \ldots, 2k-1\}$ has an expansion $W_{l^*} = \sum_i \alpha_i \Phi_i$ with at least two nonzero coefficients.*

Proof. By Lemma 3.1(i), the expansion of W_l has exactly one nonzero coefficient iff W_l equals some basis form: $W_l = \Phi_0$ iff $O(l) = \emptyset$, and $W_l = \Phi_c$ iff $O(l) = O(l_c + 1)$ (two homogeneous forms $x_0 + \sum_O t$, $x_0 + \sum_{O'} t$ agree iff $O = O'$; a relation $W_l = -\Phi_i$ or $W_l = \alpha \Phi_i$ with $\alpha \ne 1$ is impossible by the x_0-coefficient). Suppose, for contradiction, that for every l the open set $O(l)$ is $\emptyset$ or equals $O(l_c + 1)$ for some c.

Let m^* be the first slot that closes a pair α while some pair β opened after α is still open; m^* exists because π has a crossing (take the first slot at which a crossing pair closes). Set $O := O(m^* + 1)$. Then $\beta \in O$, $\alpha \notin O$, so $O \ne \emptyset$, and $m^* + 1 \le 2k - 1$ (were $m^* = 2k-1$, nothing would remain open after it). We show $O \ne O(l_c + 1)$ for every c, the desired contradiction:

- If $l_c + 1 \le m^*$: at every slot $\le m^*$ at which β is open, α is open as well (α opened before β and does not close until m^*). So if $\beta \in O(l_c + 1)$ then $\alpha \in O(l_c + 1)$, while $\alpha \notin O$; and if $\beta \notin O(l_c + 1)$ then $O(l_c + 1) \ne O$ directly. Either way $O(l_c + 1) \ne O$.
- If $l_c + 1 > m^* + 1$: the pair c belongs to $O(l_c + 1)$ and opened at $l_c > m^*$; but every pair of O is open at $m^* + 1$, hence opened before $m^* + 1$. So $c \in O(l_c + 1) \setminus O$ and $O(l_c + 1) \ne O$. (The case $l_c + 1 = m^* + 1$ cannot occur: m^* is a closing slot.)

□

Lemma 3.3 (window bound)**.** *Let $r \ge 2$, let $\alpha_1, \ldots, \alpha_r$ be nonzero integers with $\sum_i \alpha_i = 1$, let $U_1, \ldots, U_r$ be i.i.d.* Unif$[0,1]$*, and let I be any interval of length 1. Then*

$$\mathbb{P}\Big(\sum_{i=1}^{r} \alpha_i U_i \in I \Big) \ \le \ \tfrac{2}{3},$$

with equality when $r = 3$, $(\alpha_i) = (1, 1, -1)$ and $I = [0, 1]$.

Proof. If some $|\alpha_j| \geq 2$: conditionally on the other variables, $\alpha_j U_j$ is uniform on an interval of length ≥ 2, so it lands in any unit window with probability $\leq \frac{1}{2}$; integrating, the mass is $\leq \frac{1}{2}$.

If all $|\alpha_i| = 1$: r must be odd (a sum of r odd numbers has the parity of r, and it equals 1), so $r \geq 3$. Each $-U_i$ equals $U_i' - 1$ with $U_i' = 1 - U_i \sim \mathrm{Unif}[0,1]$, so $\sum \alpha_i U_i$ is a shift of an Irwin–Hall sum Σ_r of r uniforms, whose density f_r is symmetric and unimodal. For $r = 3$: the best unit window is the central one, and $\mathbb{P}(\Sigma_3 \in [1,2]) = 1 - 2\,\mathbb{P}(\Sigma_3 < 1) = 1 - \frac{2}{6} = \frac{2}{3}$ (each tail is a corner simplex of volume $1/6$); for the sign pattern $(1,1,-1)$ and $I = [0,1]$ this is attained exactly. For odd $r \geq 5$: $\sup f_r = f_r(r/2) \leq f_5(5/2) = \frac{115}{192} < \frac{2}{3}$, so any unit window has mass $< \frac{2}{3}$. (The central values decrease in r: since $f_{r+1} = f_r * \mathbf{1}_{[0,1]}$ we have $f_{r+1}\big(\frac{r+1}{2}\big) = \int_{r/2-1/2}^{r/2+1/2} f_r(x)\,dx \leq \sup f_r = f_r\big(\frac{r}{2}\big)$, the last equality by symmetry and unimodality; $f_5(5/2) = \frac{115}{192}$ is the standard Irwin–Hall evaluation.) □

Lemma 3.4 (opening-slot coordinates, Hankel)**.** *Let π be parity-mixing and $B_l := (-1)^l A_l$. Then $B_l = x_0 + \sum_{c \in O(l)} \varepsilon_c\, y_c$ with fixed signs $\varepsilon_c = -(-1)^{l_c}$; the map $(x_0, y) \mapsto (B_0, B_{l_1+1}, \ldots, B_{l_k+1})$ is unimodular; the constraint on B_l is the unit window $[0,1]$ (l even) or $[-1,0]$ (l odd); and the conclusions (iii) of Lemma 3.1 and Lemma 3.2 hold verbatim with $\Phi_c = B_{l_c+1}$ and the signed variables $\varepsilon_c y_c$ in place of t_c.*

Proof. $A_{l+1} = s_l - A_l$ gives $B_{l+1} = (-1)^{l+1}(s_l - A_l) = B_l + (-1)^{l+1} s_l$, so $B_l = x_0 - \sum_{r<l} (-1)^r s_r$; for a closed pair ($m_c < l$) the two appearances of y_c carry opposite signs $(-1)^{l_c}, (-1)^{m_c}$ and cancel *because* π is parity-mixing, leaving exactly the open pairs with the stated signs. Unimodularity is the same triangular structure as in Lemma 3.1(ii) (the diagonal entries are $\varepsilon_c = \pm 1$). The open-set argument of Lemma 3.2 uses only the combinatorics of $O(\cdot)$ and the x_0-coefficients, not the signs, so it transfers verbatim: signed open sums $x_0 + \sum_O \varepsilon y$ and $x_0 + \sum_{O'} \varepsilon y$ agree as forms iff $O = O'$ (each pair's sign is fixed once and for all). □

Theorem 3.5. *Let $\pi \in \mathcal{P}_2(2k)$ (parity-mixing in the Hankel case). If π has a crossing, then*

$$p_T(\pi) \leq \tfrac{2}{3}, \qquad p_H(\pi) \leq \tfrac{2}{3},$$

and $2/3$ is attained in the Toeplitz case (by the one-crossing pairing, arbitrarily padded with non-crossing material). Hence $p(\pi) = 1$ if and only if π is non-crossing, in both ensembles, and $\sup_{\pi \text{ crossing}} p_T(\pi) = \frac{2}{3}$.

Proof. Toeplitz. By Lemma 3.1(ii), $p_T(\pi)$ equals the volume of $\Phi(P_\pi)$. Among its defining constraints are the $k+1$ basis-cube constraints $\Phi_i \in [0,1]$ and, by Lemma 3.2, a constraint $\sum_i \alpha_i \Phi_i \in [0,1]$ whose integer coefficients are ≥ 2 in number, nonzero, and sum to 1 (Lemma 3.1(iii)). Discarding all other constraints and applying Fubini on the cube,

$$p_T(\pi) \;\leq\; \mathbb{P}\Big(\sum_i \alpha_i U_i \in [0,1]\Big) \;\leq\; \tfrac{2}{3}$$

by Lemma 3.3. *Sharpness:* for $\pi^\times = \{0,2\},\{1,3\}$ the image polytope is exactly $\{(\Phi_0, \Phi_1, \Phi_2) \in [0,1]^3 : \Phi_0 - \Phi_1 + \Phi_2 \in [0,1]\}$ (four constraints in total), whose volume is the equality case of Lemma 3.3: $p_T(\pi^\times) = \frac{2}{3}$. By Lemmas 2.3–2.4, inserting non-crossing material preserves this value, so volume-$\frac{2}{3}$ crossing pairings exist for every $k \geq 2$.

Hankel. By Lemma 3.4, $p_H(\pi)$ is the volume of a polytope whose constraints include $k+1$ unit-window constraints on the basis coordinates and one constraint $\sum_i \alpha_i B_i \in J$ (J a unit

window) with $r \geq 2$ nonzero integer coefficients. Recentring each basis coordinate to Unif$[0,1]$ translates J by an integer; Lemma 3.3 covers *arbitrary* unit windows, so $p_H(\pi) \leq \frac{2}{3}$.

In both ensembles, Corollary 2.5 gives $p = 1$ on non-crossing pairings, completing the equivalence. □

Corollary 3.6. *For all $k \geq 1$,*

$$m_{2k}(\gamma_T) \;\leq\; \mathrm{Cat}_k + \tfrac{2}{3}\big((2k-1)!! - \mathrm{Cat}_k\big), \qquad m_{2k}(\gamma_H) \;\leq\; \mathrm{Cat}_k + \tfrac{2}{3}\big(k! - \mathrm{Cat}_k\big),$$

and (by Corollary 2.5) $m_{2k} \geq \mathrm{Cat}_k$ in both ensembles.

Remark 3.7. The Toeplitz bound is sharp pairing-by-pairing (Theorem 3.5). For Hankel the largest crossing volume in all computed data (through order 14, i.e. all $7! = 5040$ parity-mixing pairings and everything below) is $\frac{1}{2}$; we conjecture $\sup_{\pi \text{ crossing}} p_H(\pi) = \frac{1}{2}$ — the parity-mixing constraint excludes the $r = 3$ centred-window configuration that achieves $2/3$ in the Toeplitz case (the one-crossing pairing $\{0,2\},\{1,3\}$ is not parity-mixing).

4 Exact computation and verification

4.1 Algorithm

For each dihedral orbit representative (Lemma 2.3; e.g. 10395 pairings reduce to 554 orbits at $2k = 12$) we evaluate $\mathrm{vol}(P_\pi)$ exactly: (i) the $\leq 4k$ halfspaces (integer data) are deduplicated; (ii) vertices are enumerated by incremental *double description* over $\mathbb{Q}$, starting from the bounding box, with the standard combinatorial adjacency criterion; (iii) the polytope is triangulated combinatorially: faces are identified with their sets of incident vertices via exact tight-set masks, facets of a face are detected by exact affine-rank tests, and each face is coned from its minimal vertex; (iv) the volume is $\sum |\det|/d!$ over the resulting simplices, in exact rational arithmetic. The implementation is ∼300 lines of dependency-free Python (`exact_volume.py`); $2k = 12$ (Toeplitz) takes under two minutes on 14 cores, and the largest run — $2k = 14$, Toeplitz: 135135 pairings in 5283 dihedral orbits — about 90 minutes.

4.2 Verification

Because our value differs from a published one, we checked the pipeline in several independent ways:

1. *Published values.* The engine reproduces exactly $m_4(\gamma_T) = 8/3$, $m_6(\gamma_T) = 11$ (including Hammond–Miller's configuration data $2, 6, 3, 3, 1\times$ volumes $1, \frac{2}{3}, 1, \frac{1}{2}, \frac{1}{2}$), $m_4(\gamma_H) = 2$, $m_6(\gamma_H) = \frac{11}{2}$, and $m_8(\gamma_H) = \frac{281}{15}$ [1, 2].

2. *Independent vertex enumeration.* For every orbit at order 8 (both ensembles), brute-force enumeration over all $\binom{m}{d}$ constraint subsets reproduces the DD vertex sets exactly, and the float convex-hull volume (`scipy`/Qhull, an independent triangulation) matches each exact volume to 10^{-10}.

3. *Per-orbit Monte Carlo.* Every orbit volume at every computed order was checked by vectorized rejection sampling (10^6; at order 8, 10^8 samples per orbit): all z-scores lie in $[-2, 2]$. For the disputed total this gives $m_8(\gamma_T) = 60.544 \pm 0.0086$, i.e. the published $964/15 = 64.267$ is rejected at 432σ while $908/15 = 60.533$ agrees at 1.2σ.

4. *Eigenvalue simulation.* Empirical spectral moments of simulated matrices ($n = 1000, 2000, 4000$; 320/96/28 repetitions) extrapolated linearly in $1/n$ agree with every exact value; e.g. $m_8(\gamma_T)$: 60.87 ± 2.51; $m_{10}(\gamma_T)$: 416 ± 34 (exact 415); $m_{12}(\gamma_T)$: 3384 ± 535 (exact 3405.7); $m_{12}(\gamma_H)$: 332 ± 18 (exact 350.7, 1σ); $m_{10}(\gamma_H)$: 74.4 ± 2.3 (exact 75.47).

5. *Independent exact counting.* For Rademacher entries, $\mathbb{E}\,\mathrm{tr}(A_n/\sqrt{n})^{2k}/n$ is an exactly computable integer count (circuits whose slot-value multiset has all even multiplicities), evaluated by a meet-in-the-middle dynamic program with multiset parity tracked through 127-bit hashing, and is an Ehrhart-type quasi-polynomial in n. For Hankel, interpolation through $n \in \{4, \ldots, 14\}$ with held-out exact prediction at $n = 16, 18$ recovers $m_6(\gamma_H) = \frac{11}{2}$ and $m_8(\gamma_H) = \frac{281}{15}$ *exactly* — validating the counting route — while the Toeplitz counts converge too slowly at these n for naive interpolation; a three-term fit of the Toeplitz counts at $n = 14, 16, 18$ extrapolates to ≈ 61.5, consistent with 908/15 and not with 964/15.

6. *Internal consistency.* Orbit-constancy of volumes (Lemma 2.3) was verified on random orbit members at every order; the volume-1 pairings in every dataset are exactly the non-crossing ones, Cat_k in number (Theorem 3.5); brute-force summation over all 105 pairings at $2k = 8$ (no symmetry reduction) reproduces 908/15.

These checks differ in logical strength. Items 1–4 and 6 bear directly on the disputed value: two exact independent computations of the same polytope data (double description vs. brute-force basis enumeration), an independent float triangulation, a 432σ Monte Carlo discrimination, and direct eigenvalue simulation. Item 5 is an *exact* independent route where it terminates (the Hankel controls at orders 6 and 8) and a consistency check — not an exact verification — for the Toeplitz value, whose finite-n counts converge too slowly for interpolation at accessible sizes.

5 Results

Theorem 5.1. *The even moments of the Toeplitz and Hankel limiting spectral distributions are*

$2k$	$m_{2k}(\gamma_T)$	*numeric*	$m_{2k}(\gamma_H)$	*numeric*
2	1	1	1	1
4	8/3	2.6667	2	2
6	11	11	11/2	5.5
8	**908/15**	60.5333	281/15	18.7333
10	**415**	415	**2717/36**	75.4722
12	**23840/7**	3405.7143	**1052/3**	350.6667
14	**325719/10**	32571.9	**1331087/720**	1848.7319

Bold values are new (the bold m_8 corrects [2]); odd moments vanish.

5.1 Volume spectra

The volumes take few distinct rational values, shared across ensembles (consequences of Lemmas 2.3–2.4: padding a "core" crossing diagram with non-crossing material preserves its volume).

Toeplitz.

$2k$	volume × number of pairings
4	1×2; $\frac{2}{3} \times 1$
6	1×5; $\frac{2}{3} \times 6$; $\frac{1}{2} \times 4$
8	1×14; $\frac{2}{3} \times 28$; $\frac{1}{2} \times 32$; $\frac{9}{20} \times 4$; $\frac{2}{5} \times 5$; $\frac{11}{30} \times 22$
10	1×42; $\frac{2}{3} \times 120$; $\frac{1}{2} \times 180$; $\frac{9}{20} \times 40$; $\frac{4}{9} \times 5$; $\frac{2}{5} \times 50$; $\frac{11}{30} \times 220$; $\frac{61}{180} \times 40$; $\frac{1}{3} \times 6$; $\frac{13}{45} \times 70$; $\frac{49}{180} \times 150$; $\frac{1}{4} \times 22$

(the $2k = 12$ spectrum, with 554 orbits and 28 distinct volumes, is in Appendix A).

Hankel.

$2k$	volume × number of pairings
4	1×2
6	1×5; $\frac{1}{2} \times 1$
8	1×14; $\frac{1}{2} \times 8$; $\frac{11}{30} \times 2$
10	1×42; $\frac{1}{2} \times 45$; $\frac{11}{30} \times 20$; $\frac{1}{3} \times 1$; $\frac{13}{45} \times 5$; $\frac{49}{180} \times 5$; $\frac{1}{4} \times 2$
12	1×132; $\frac{1}{2} \times 220$; $\frac{11}{30} \times 132$; $\frac{1}{3} \times 12$; $\frac{13}{45} \times 60$; $\frac{49}{180} \times 60$; $\frac{23}{90} \times 6$; $\frac{1}{4} \times 24$; $\frac{5}{21} \times 6$; $\frac{68}{315} \times 12$; $\frac{29}{140} \times 2$; $\frac{509}{2520} \times 36$; $\frac{239}{1260} \times 12$

Note the volume-1 counts $2, 5, 14, 42, 132$: the Catalan numbers, as proved in Theorem 3.5.

5.2 The eighth Toeplitz moment: correction of Hammond–Miller

At $2k = 8$ the 105 pairings split into 17 dihedral orbits carrying six distinct volumes; grouping by volume:

$$m_8(\gamma_T) = 14 \cdot 1 + 28 \cdot \tfrac{2}{3} + 32 \cdot \tfrac{1}{2} + 4 \cdot \tfrac{9}{20} + 5 \cdot \tfrac{2}{5} + 22 \cdot \tfrac{11}{30} = 14 + \tfrac{56}{3} + 16 + \tfrac{9}{5} + 2 + \tfrac{121}{15} = \tfrac{908}{15},$$

i.e. $60\frac{8}{15}$, in place of the $64\frac{4}{15}$ of [2]. The multiplicities above sum to 105, and every individual volume is verified by the methods of Section 4.2. All other values of [2], including $m_6 = 11$ with its configuration table, are confirmed by our computation.

6 Consequences and observations

6.1 Gaussian and factorial deficits

Write $\delta_k^T = m_{2k}(\gamma_T)/(2k-1)!!$ (Gaussian deficit) and $\delta_k^H = m_{2k}(\gamma_H)/k!$ (deficit against the symmetrized-Rayleigh/reverse-circulant moments $k!$):

$$\delta^T : \ 1, \ \tfrac{8}{9}, \ \tfrac{11}{15}, \ \tfrac{908}{1575}, \ \tfrac{83}{189}, \ \tfrac{4768}{14553}, \ \tfrac{36191}{150150} \qquad \delta^H : \ 1, \ 1, \ \tfrac{11}{12}, \ \tfrac{281}{360}, \ \tfrac{2717}{4320}, \ \tfrac{263}{540}, \ \tfrac{1331087}{3628800}.$$

Both decrease strictly from $k = 2$ resp. 3 on (consistent with Theorem 3.5); the successive ratios δ_k/δ_{k-1} are

Toeplitz ($k = 2..7$): .8889, .8250, .7861, .7617, .7461, .7357, Hankel ($k = 3..7$): .9167, .8515, .8058, .7744, .7

6.2 Numerical observations: deficit ratios and norm constants

Everything in this subsection is heuristic extrapolation from the exact data; nothing here is proved, and none of the results elsewhere in the paper depend on it. If $m_{2k} \approx C\,(2k-1)!!\,\sigma_*^{2k}$ for large k — the moment growth of a measure with Gaussian-type upper tail of scale σ_* — then $\delta_k^T/\delta_{k-1}^T \to \sigma_*^2$. Sen and Virág [9] proved $\|T_n\|/\sqrt{2n\log n} \to \sigma_{SV} = 0.8288996\ldots$, corresponding to a Gaussian-type tail of scale σ_{SV} for γ_T. The exact Toeplitz ratios above, extrapolated in $1/k$ (linear through the last two: 0.674; quadratic through the last three: 0.688), are consistent with $\sigma_{SV}^2 = 0.68707\ldots$, approaching it from below with the slow convergence expected of a $1/k$-corrected sequence.

For Hankel the correct comparison scale is factorial: if $m_{2k} \approx C\,k!\,\theta^k$ — the growth of a symmetrized-Rayleigh-type law with density $\asymp |x|e^{-x^2/\theta}$, cf. the reverse-circulant limit where $m_{2k} = k!$ exactly ($\theta = 1$) — then $\delta_k^H/\delta_{k-1}^H \to \theta$ and the norm constant heuristic gives $\|H_n\|/\sqrt{2n\log n} \to \sqrt{\theta/2}$ (for the reverse circulant this gives $1/\sqrt{2}$, which is the known value). Our exact ratios extrapolate to $\theta \approx 0.63$–0.65, yielding the *data-driven prediction*

$$\lim_{n\to\infty} \frac{\|H_n\|}{\sqrt{2n\log n}} \;\approx\; 0.56\text{–}0.57,$$

for the (open) Hankel norm constant. Direct simulation corroborates the range: $\|H_n\|/\sqrt{2n\log n} = 0.655, 0.657, 0.627, 0.617$ at $n = 10^3, 2{\cdot}10^3, 4{\cdot}10^3, 8{\cdot}10^3$, still decreasing — and the same estimator at $n = 4000$ overshoots the *proven* Toeplitz constant by a comparable margin (0.839 vs. 0.8289), as expected for $\sqrt{\log n}$-type convergence. The underlying heuristic — polynomially many weakly dependent local maxima realize the extreme-value scale predicted by independence — is robust in random matrix theory; see [10] for a sharp instance of the same principle, where even *exponentially* many strongly dependent submatrix statistics exhibit the independence prediction for their extreme, established by the second-moment method. We state the Hankel value as data; a proof is beyond the scope of this paper.

6.3 Tail bounds

By Markov's inequality applied to the new top moments (optimizing over the available orders), for all $x > 0$:

$$\gamma_T([x,\infty)) \le \tfrac{1}{2}\min_{k\le 7} \frac{m_{2k}(\gamma_T)}{x^{2k}}, \qquad \gamma_H([x,\infty)) \le \tfrac{1}{2}\min_{k\le 7} \frac{m_{2k}(\gamma_H)}{x^{2k}},$$

e.g. $\gamma_T([3,\infty)) \le 3.2\times 10^{-3}$ and $\gamma_H([3,\infty)) \le 1.9\times 10^{-4}$, sharpening the order-8 bounds (4.6×10^{-3}, resp. 1.4×10^{-3}); at $x = 4$ the new bounds give 6.1×10^{-5} and 3.4×10^{-6}.

6.4 Candidate laws: exclusions and a certified approximation

The exact moments through order 12 (resp. 14) sharply constrain any candidate closed form. Three natural families:

- *Gaussian scale mixtures* $X = WG$ ($W \ge 0$ independent of G): these have $m_4 \ge 3m_2^2 = 3$; since $m_4(\gamma_T) = \frac{8}{3} < 3$ and $m_4(\gamma_H) = 2 < 3$, *neither limit law is any Gaussian scale mixture* (so neither is a "randomized-variance Gaussian").

- *Symmetrized gamma* ($f \propto |x|^{a-1}e^{-|x|/s}$): fitting (m_2, m_4) forces $a = 2.72$ (T), 4.37 (H); the predicted m_6 already misses by 24% resp. 22%, and by order 12 the discrepancy exceeds 500%. Excluded.

- *Generalized error distributions* $f \propto e^{-|x/\alpha|^\beta}$: fitting (m_2, m_4) for Toeplitz gives $\beta = 2.4367$, and then the GED predicts $m_6 = 11.036$, $m_8 = 60.762$, $m_{10} = 413.36$: within 0.4% of the exact values through order ten, drifting to 2.3% at order twelve and 5.5% at order fourteen. Thus γ_T is *provably not* a GED (the exact $m_6 = 11 \neq 11.036$ settles it), yet a GED with $\beta \approx 2.44$ is an accurate two-parameter surrogate, now with certified error bounds at every order up to 12 — a practical approximation that order-≤ 8 data could neither calibrate nor falsify. For Hankel the fitted GED ($\beta = 6$) fails immediately (6% at m_6, 47% at m_{12}): γ_H is far from any GED.

6.5 Toward identifying the laws

The full volume spectra, not just their sums, are intrinsic invariants of the combinatorial structure and may guide a future exact solution; the repeated values across T and H (e.g. $\frac{11}{30}, \frac{13}{45}, \frac{49}{180}$) reflect shared crossing "cores" (Lemmas 2.3–2.4 imply a core's volume is invariant under non-crossing padding and dihedral moves), and suggest a calculus of core diagrams worth developing.

The volume-1 class itself carries algebraic structure: the non-crossing pairings of $2k$ points are exactly the diagram basis of the Temperley–Lieb algebra $\mathrm{TL}_k(\delta)$, and the repository accompanying this paper contains an independent enumeration of that basis (with the TL relations verified at $\delta = 2$, where stacking diagrams weights each closed loop by 2) ported from a diagrammatic R program of the author (`references/`); the basis sizes $2, 5, 14, 42, 132, 429$ reproduce the volume-1 counts of the spectra above, an independent combinatorial confirmation of Theorem 3.5's "only if" direction across all computed data. Whether the crossing-volume function $\pi \mapsto p(\pi)$ factors through a deformation of the Temperley–Lieb quotient (crossings resolved with weights, as in skein calculus) is an intriguing question: the small denominators and heavy repetition in the spectra suggest that a finite "core skein" calculus might compute all volumes.

7 Reproducibility

All code (pure Python: `tk_core.py`, `exact_volume.py`, `run_moments.py`, every verification script) and all data (machine-readable per-orbit exact volumes as JSON) are public at

`https://github.com/yspennstate/toeplitz-hankel-exact-moments`

together with scripts that regenerate every table of this paper from scratch. Every number in Theorem 5.1 is reproducible end-to-end on a laptop; the values are exact rationals produced by integer-only arithmetic, and the disputed order ($2k = 8$, Toeplitz) is covered by a second, independent exact implementation (`brute_vertices_check.py`) sharing no code with the first.

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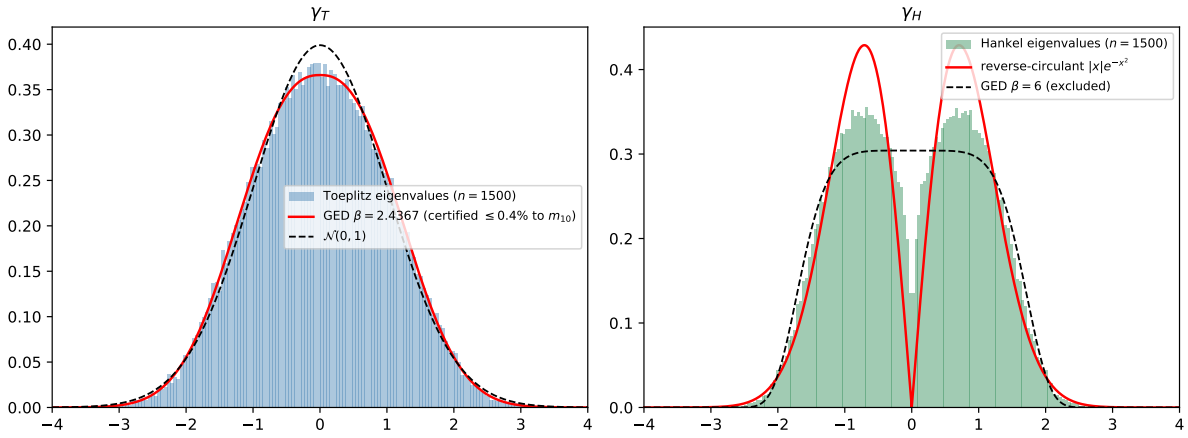


Figure 1: Simulated spectral densities ($n = 1500$, 60 repetitions). Left: γ_T with the GED surrogate ($\beta = 2.4367$; indistinguishable at this resolution, provably distinct at order 6) and the standard Gaussian. Right: γ_H (bimodal) with the reverse-circulant density $|x|e^{-x^2}$ and the excluded GED fit.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [11] and working together inside the Brain framework [13], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [12], the Math Brain repository [11] and the Brain framework repository [13].

Code and data availability

The code and data for this paper are at `https://github.com/yspennstate/toeplitz-hankel-exact-moments`. The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [11].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools (large language model assistants) to assist with drafting, revision and bibliographic organization. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the manuscript.

Declaration of competing interest

The author declares no competing interests.

A Complete volume spectra at the highest orders

Toeplitz, $2k = 12$ (10395 pairings, 554 dihedral orbits, 28 distinct volumes; the $132 = \mathrm{Cat}_6$ non-crossing pairings carry volume 1):

$$1 \times 132;\ \tfrac{2}{3} \times 495;\ \tfrac{1}{2} \times 880;\ \tfrac{9}{20} \times 264;\ \tfrac{4}{9} \times 66;\ \tfrac{2}{5} \times 330;\ \tfrac{11}{30} \times 1452;\ \tfrac{61}{180} \times 480;\ \tfrac{1}{3} \times 120;\ \tfrac{43}{140} \times 4;\ \tfrac{383}{1260} \times 18;$$
$$\tfrac{13}{45} \times 840;\ \tfrac{2}{7} \times 7;\ \tfrac{49}{180} \times 1800;\ \tfrac{19}{70} \times 60;\ \tfrac{23}{90} \times 96;\ \tfrac{1}{4} \times 396;\ \tfrac{313}{1260} \times 132;\ \tfrac{5}{21} \times 102;\ \tfrac{71}{315} \times 54;\ \tfrac{68}{315} \times 456;$$
$$\tfrac{19}{90} \times 234;\ \tfrac{22}{105} \times 117;\ \tfrac{29}{140} \times 80;\ \tfrac{17}{84} \times 82;\ \tfrac{509}{2520} \times 984;\ \tfrac{239}{1260} \times 528;\ \tfrac{19}{105} \times 186.$$

Sum: $m_{12}(\gamma_T) = 23840/7$.

Hankel, $2k = 14$ (5040 pairings, 260 orbits, 39 distinct volumes; the $429 = \mathrm{Cat}_7$ non-crossing pairings carry volume 1):

$$1 \times 429;\ \tfrac{1}{2} \times 1001;\ \tfrac{11}{30} \times 728;\ \tfrac{1}{3} \times 91;\ \tfrac{13}{45} \times 455;\ \tfrac{49}{180} \times 455;\ \tfrac{23}{90} \times 84;\ \tfrac{1}{4} \times 190;\ \tfrac{5}{21} \times 84;\ \tfrac{68}{315} \times 168;\ \tfrac{29}{140} \times 28;$$
$$\tfrac{17}{84} \times 7;\ \tfrac{509}{2520} \times 504;\ \tfrac{239}{1260} \times 168;\ \tfrac{17}{90} \times 14;\ \tfrac{59}{315} \times 14;\ \tfrac{31}{168} \times 7;\ \tfrac{19}{105} \times 84;\ \tfrac{5}{28} \times 14;\ \tfrac{433}{2520} \times 7;\ \tfrac{1}{6} \times 7;\ \tfrac{167}{1008} \times 14;$$
$$\tfrac{59}{360} \times 7;\ \tfrac{809}{5040} \times 7;\ \tfrac{1613}{10080} \times 28;\ \tfrac{1591}{10080} \times 56;\ \tfrac{197}{1260} \times 28;\ \tfrac{781}{5040} \times 7;\ \tfrac{16}{105} \times 28;\ \tfrac{253}{1680} \times 14;\ \tfrac{379}{2520} \times 42;\ \tfrac{1511}{10080} \times 42;$$
$$\tfrac{367}{2520} \times 14;\ \tfrac{241}{1680} \times 42;\ \tfrac{353}{2520} \times 56;\ \tfrac{349}{2520} \times 56;\ \tfrac{683}{5040} \times 14;\ \tfrac{47}{360} \times 32;\ \tfrac{109}{840} \times 14.$$

Sum: $m_{14}(\gamma_H) = 1331087/720$. Note the leading segments of the two spectra agree (the shared "core" phenomenon of Section 6).

Toeplitz, $2k = 14$ (135135 pairings, 5283 dihedral orbits, 70 distinct volumes; the $429 = \mathrm{Cat}_7$ non-crossing pairings carry volume 1; largest crossing volume again $\frac{2}{3} \times 2002$): the full spectrum is machine-readable in the repository (`results/volumes_T14.json`); its sum is $m_{14}(\gamma_T) = 325719/10$.

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